

Design Document: Mach-Zehnder Interferometer

SHALMOLI GHOSH

shalmolighosh@ku.edu

Abstract: This report describes the design of a Mach-Zehnder interferometer for silicon-on-insulator fabrication, considering waveguides in the strip configuration. Simulated results and analytical evaluations are presented to support design choices. A first draft of the layout mask is presented for future fabrication and testing processes.

© 2025 Optica Publishing Group

1. Introduction

The Mach-Zehnder interferometer (MZI) is one of the oldest optical instruments, in use for more than a century. The Mach-Zehnder interferometer consists of an input split into two branches (here, the upper and lower waveguides), then recombined. The splitting/combining can be achieved using any splitter/combiner, including Y-branches and directional couplers. They are widely used as high-speed modulators in telecommunications, as well as in optical switches, filters, and highly sensitive sensors.

2. Theory: MZI

In its simplest way, a balanced MZI can be modeled as a 2-port device, formed by a beam splitter and a beam-combiner, connected by waveguides with identical geometries and material properties. With this configuration, for any wavelength, the signals of the two branches are constructively combined, in the MZI output.

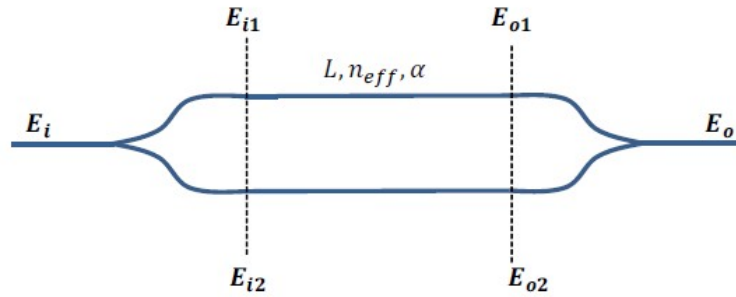


Fig. 1. MZI in a balanced configuration. Both branches have the same length L , effective index n_{eff} and propagation loss α

In order to exploit the different levels of interference, a length asymmetry can be included between the branches. In addition to that, the two waveguides can present differences in their effective indexes (n_{eff1} , n_{eff2}) and attenuation constants (α_1 , α_2). Thus, depending on the signal wavelength, the phase mismatch between the two combined signals causes different kinds of interference, from the complete constructive to the total destructive ones.

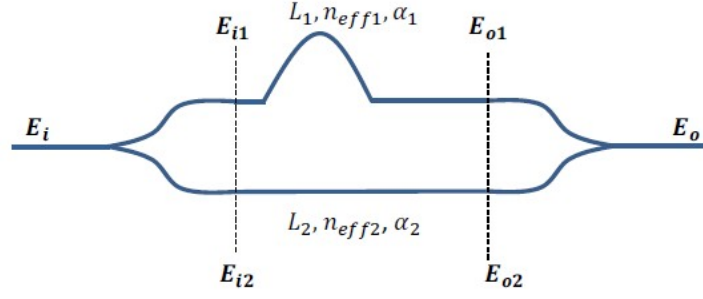


Fig. 2. MZI in an unbalanced configuration. Upper and lower branches may have distinct length and material properties.

Consider the configuration presented in Figure 2. The vectorial electrical field at the input of the MZI is represented by E_i . At the output of the beam splitter, the complex electrical fields at the upper and lower branches are given by

$$E_{i1} = E_{i2} = \frac{1}{\sqrt{2}} E_i. \quad (1)$$

Considering the propagation between the Y-branches, the electrical fields at the input of the beam-combiner can be described by

$$E_{o1} = E_{i1} \cdot e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \quad (2)$$

$$E_{o2} = E_{i2} \cdot e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \quad (3)$$

where $\beta_{1,2} = (2\pi/\lambda)n_{\text{eff}1,2}$ accounts for the propagation constants of the upper/lower branches. For the beam-combiner, each input field has the amplitude equally divided between the fundamental mode and the radiation modes [5]. Thus,

$$E_o = \frac{1}{\sqrt{2}} \cdot (E_{o1} + E_{o2}). \quad (4)$$

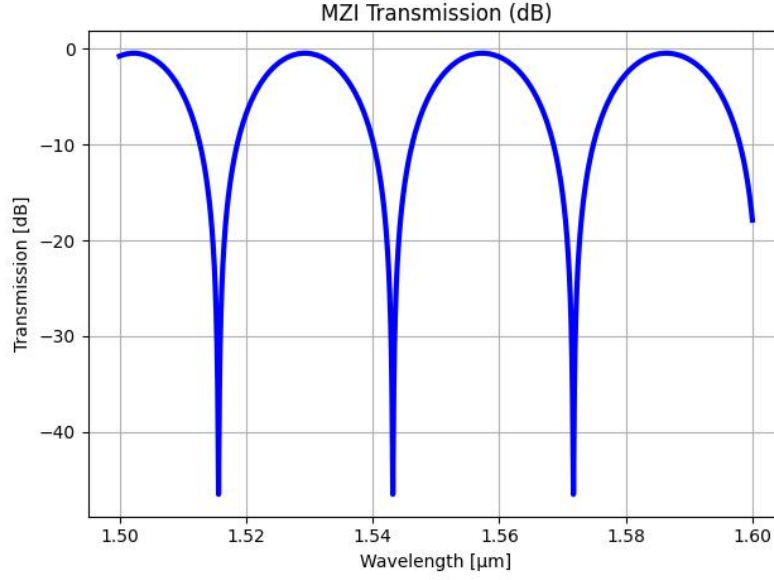


Fig. 3. Example of an unbalanced MZI transfer function with a FSR of 28nm

Substituting Equations (1)-(3) in Equation (4), we find that

$$E_o = \frac{1}{2} \cdot E_i \left(e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right). \quad (5)$$

As the light intensity is proportional to the squared power of the field amplitude, the relation between the light intensity in the output/input of the MZI is

$$I_o = \frac{1}{4} \cdot I_i \left| \left(e^{-j\beta L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta L_2 - \frac{\alpha_2}{2} L_2} \right) \right|^2 \quad (6)$$

Equation (6) represents the general form of an unbalanced MZI transfer function. However, for the purpose of this work, we can neglect the propagation loss relative to the constants α_1 and α_2 . In addition to that, the propagation constants may also be considered equivalent in both branches. In this way, Equation (6) may be simplified to

$$I_o = \frac{1}{2} \cdot I_i (1 + \cos(\beta \Delta L)) \quad (7)$$

where $\Delta L = L_1 - L_2$, and $\beta_1 = \beta_2 = \beta$. Figure 4 exemplifies the unbalanced MZI transfer function, considering a path unbalance of 20 μm , for the TE mode. The period between the MZI transfer function peaks, i.e., the free spectral range (FSR), is calculated by [5]

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L \cdot n_g(\lambda)}, \quad (8)$$

where n_g is the group index, defined by [5]

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \cdot \frac{dn_{eff}(\lambda)}{d\lambda}. \quad (9)$$

3. Modelling and Simulation

In our design, we use a strip waveguide with a width of 500 nm and a height of 220 nm. The simulated waveguide mode profiles are presented in the following figures.

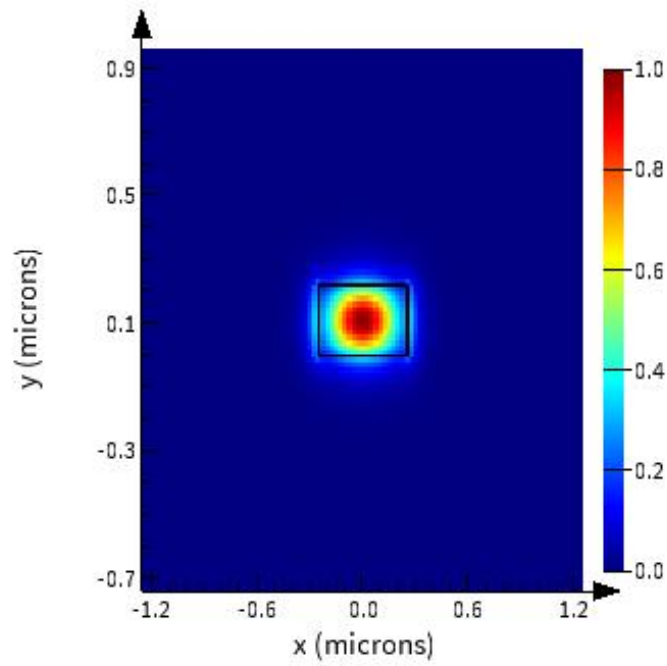


Fig. 4. Electric field intensity of TE mode in the waveguide

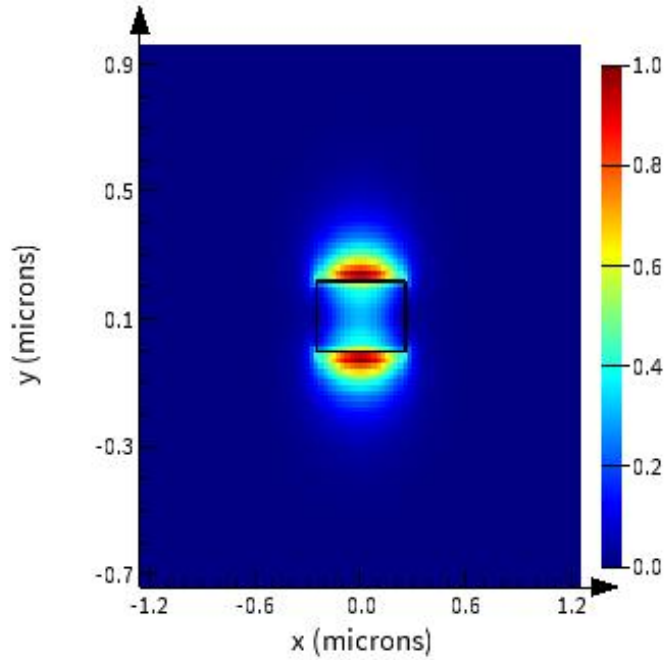


Fig. 5. Electric field intensity of TM mode in the waveguide

The following table shows the effective index for different modes:

Table 1. EFFECTIVE INDEXES CORRESPONDING TO THE GUIDED MODES (1500NM) OF THE SIMULATED STRIP WAVEGUIDE (220NMX500NM)

Mode	Effective Index
TE1	2.5011
TM1	1.8391
TE2	1.5309
TM2	1.3660

For the aforementioned geometry, a modal analysis simulation was performed with the Lumerical MODE software [6], for a wavelength of 1500 nm. The guided modes, with their corresponding effective indexes, are listed in Table I. The imaginary part of each index was neglected due to the very low loss-level. So, the modeling will be conducted for the first (quasi-) TE mode. Its mode profile (electrical field intensity) is shown in Figure 4. This figure also highlights, with a black solid line, the waveguide region. Next, a wavelength sweep was performed in the interval between 1500nm and 1600nm. This simulation accounted for material and waveguide dispersions. The results, in terms of group index and effective index, are presented in Figures 6 and 7. It should be noted that these two parameters are fundamental for the design of any device.

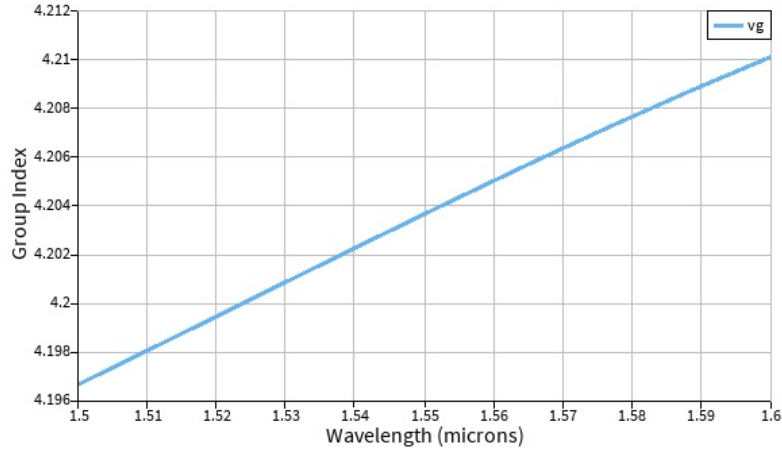


Fig. 6. Group index versus wavelength, for the TE1 mode.

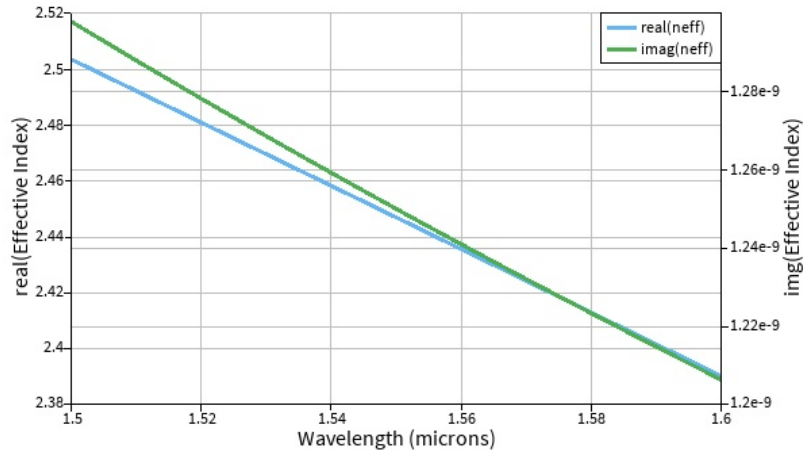


Fig. 7. Effective index versus wavelength, for the TE1 mode.

After obtaining the modal analysis data, the effective index data was fit with respect to the wavelength. Considering a truncated Taylor-series (with three coefficients), the effective index can be modelled as [5]

$$n_{eff}(\lambda[\mu m]) = n_{eff1} + n_{eff2}(\lambda - 1.5) + n_{eff3}(\lambda - 1.5)^2. \quad (10)$$

Figure 8 shows the fit result, where the obtained polynomial expression was

$$n_{eff}(\lambda[\mu m]) = 2.5032 - 1.129(\lambda - 1.5) + -0.439(\lambda - 1.5)^2. \quad (11)$$

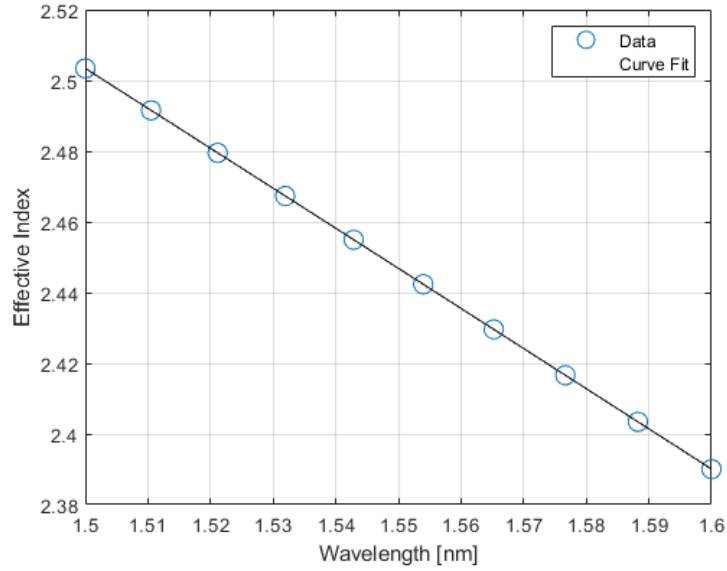


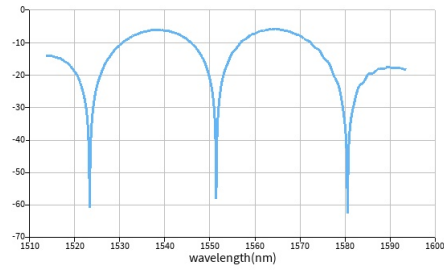
Fig. 8. Compact model for the waveguide mode (TE1).

4. MZI Analysis

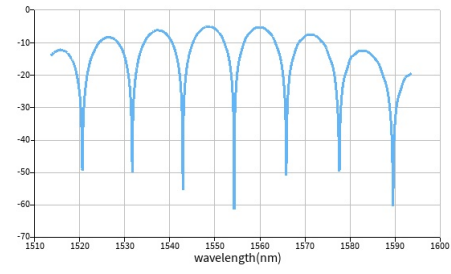
We simulated the MZI using Lumerical INTERCONNECT and get the spectrum of the interferometer. We choose TE mode in this simulation and the path length difference is 100 μm . To perform the light coupling, grating couplers were included in the input/output of the model. As described in Section 2, the unbalanced MZI consists of two Y branches (a splitter and a combiner) connected by waveguides (upper and lower branches). In this work, the interference at the MZI output is exploited only by the length mismatch between the upper and lower branches. Therefore, both waveguides were modelled for the same cross-section geometry and material properties. As per equation [8], the FSR is calculated and the FSR with different ΔL in our design is shown in Table 2.

Table 2. FSR WITH DIFFERENT ΔL

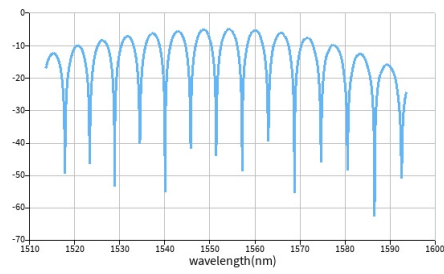
$\Delta L(\mu\text{m})$	FSR(nm)
20	28.26
50	11.31
100	5.65
150	3.77
200	2.83



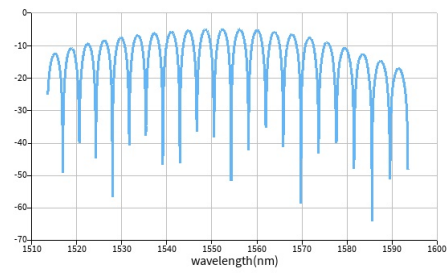
(a)



(b)



(c)



(d)

Fig. 9. Transfer function of several MZIs, with ΔL : a) 20 μm , b) 50 μm , c) 100 μm and d) 150 μm .

As predicted by Equation (8), the FSR decreases with the length unbalance. Figure 9 presents the transfer functions of the MZI with ΔL varying from 20 to 150 μm .

5. Simulation Results

The gratings and Y-branches were described on the basis of S parameters provided in the devices library of this course [8].

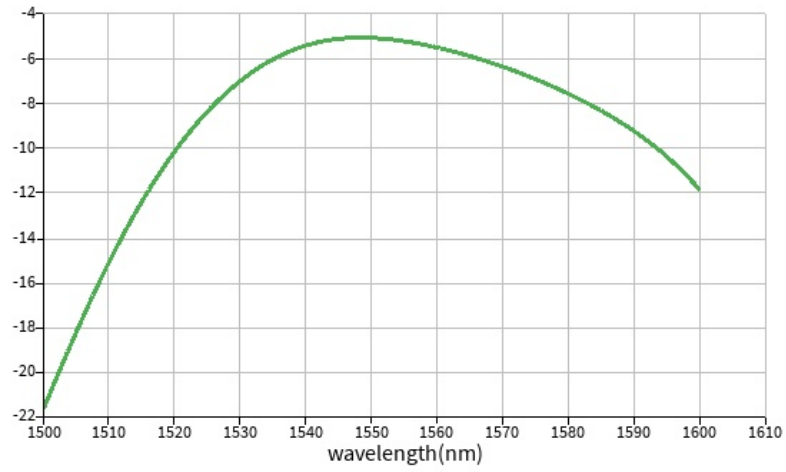


Fig. 10. Insertion loss of two grating couplers. Y-axis in dB.

The insertion losses of the grating couplers, depicted in Figure 9, show insertion losses ranging from 5dB to 22dB.

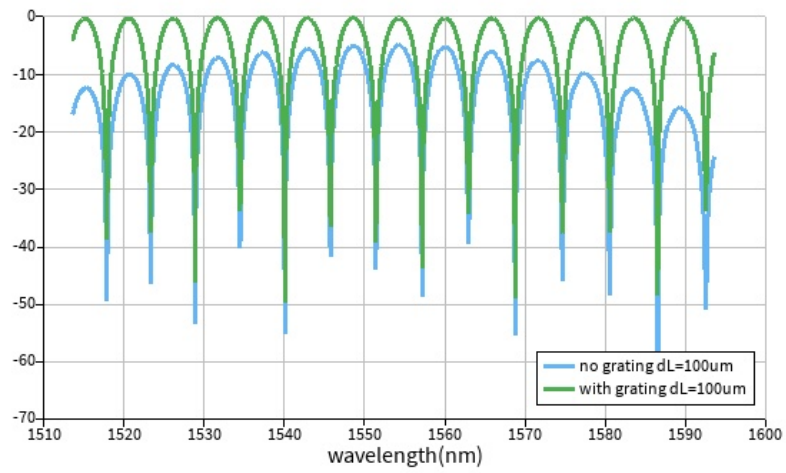


Fig. 11. Spectrum of MZI with $\Delta L = 100\mu\text{m}$.

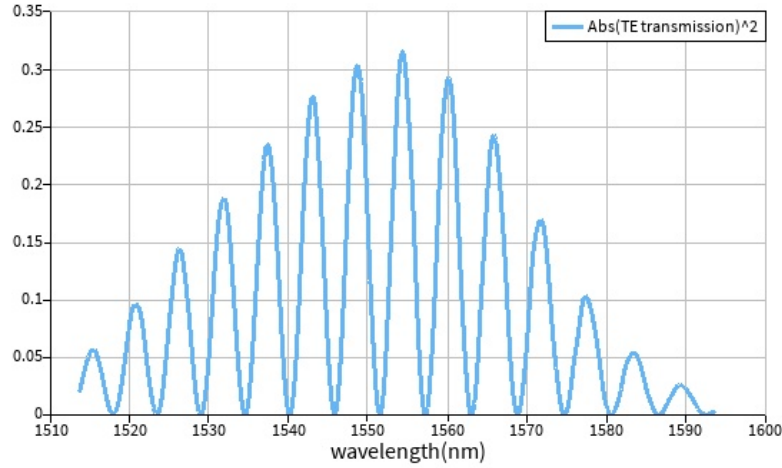


Fig. 12. Transmission of MZI with two grating couplers.

Till now, all designs were projected only for the first TE mode. The relation between the FSR and the MZI geometry was investigated, based on theoretical and simulation models. For the layout, these analyzed models were converted in a layout description, for future fabrication and measurement stages. The test data will be used for the validation of the posterior design and the conclusion of the MZI analysis.

6. Layout Design

The first design proposal was developed in KLayout software. The layout mask is depicted in Figure 13, for different MZIs, for which the transmission vs wavelength results were already summarized in Figure 14. A few details were demanded in design, in order to make possible the future automated testing. The main considered restrictions were the following: first, the TE grating couplers were oriented to the right side, to speed up the tests, avoiding unnecessary chip rotations; second, a 127 μm pitch was fixed, for the input and output gratings, to match with the pitch of the fiber array used in the automated measurements. The mask view of the proposed designs is illustrated in Figure 13.

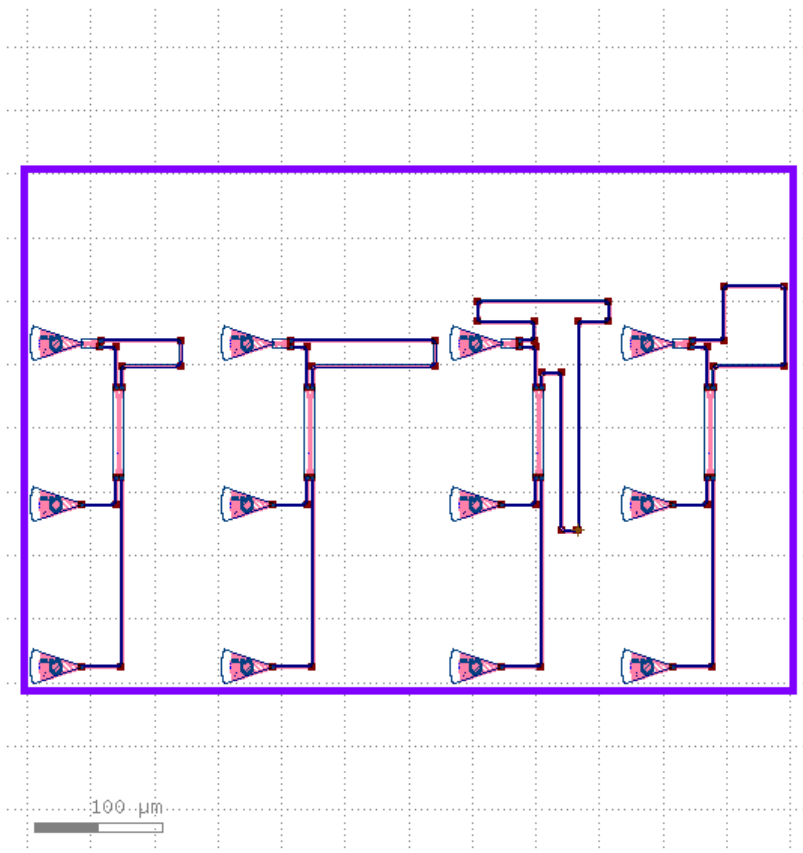


Fig. 13. Layout of the MZI devices.

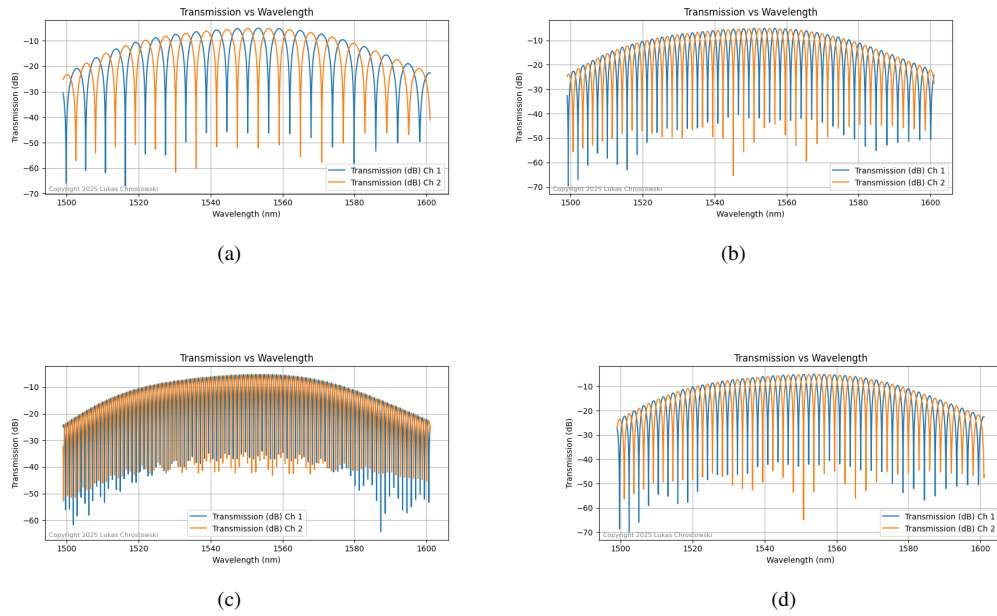


Fig. 14. Transmission spectrum of the MZIs from INTERCONNECT, with ΔL : a) 100 μm , b) 200 μm , c) 500 μm and d) 202 μm (different variant of MZI2).

After the first design draft review, added a loopback grating coupler structure in a new layout for loopback calibration. Also added some different MZIs in the second version of the layout. The layout mask is depicted in Figure 15, for different MZIs and Grating Coupler, and the transmission vs wavelength results were summarized in Figure 16.

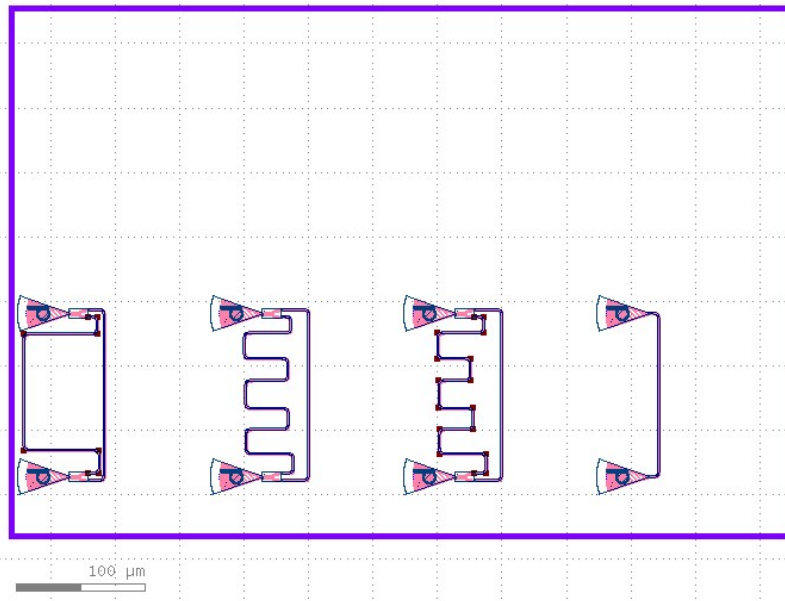


Fig. 15. Second version of the Layout of the MZI devices.

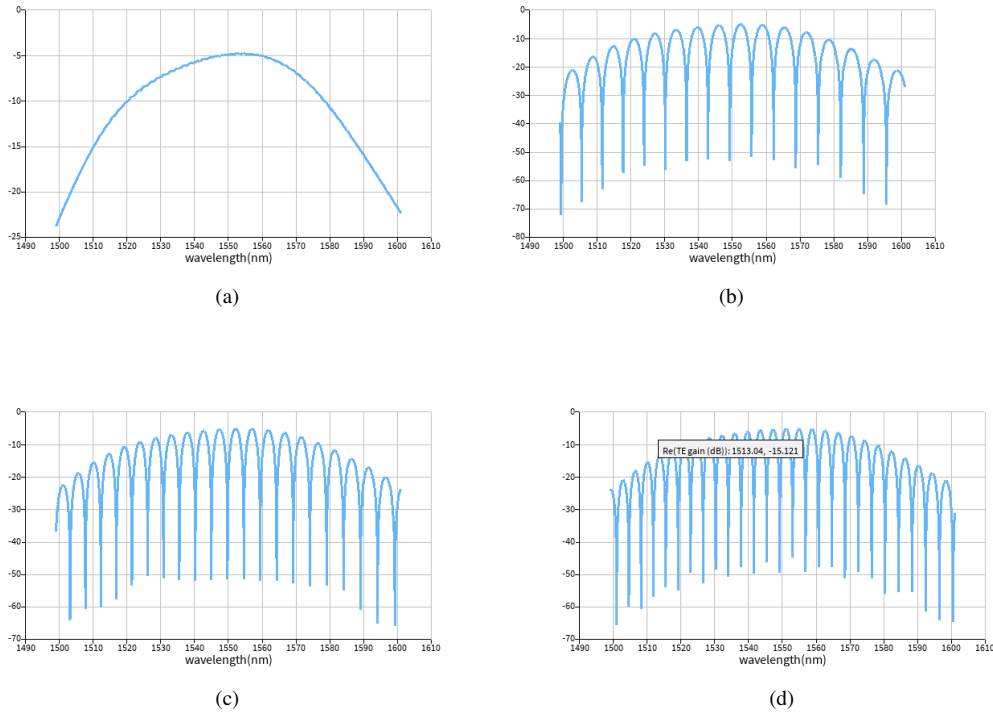


Fig. 16. Transmission spectrum of the a) GC and MZIs from INTERCONNECT, with ΔL : b) $89 \mu\text{m}$, c) $120 \mu\text{m}$, d) $147 \mu\text{m}$.

7. Design variability

Due to fabrication issues, there are several aspects that include mismatches between the designed and the fabricated devices. Mainly, these manufacturing errors generate variabilities on the thickness and width of the waveguides.

As an approximation, it is usual to associate the fabrication variability of a given geometrical parameter with a Gaussian distribution, where the nominal value corresponds to the distribution mean (μ), and the fabrication error is expressed by the standard deviation (σ).

Regarding the waveguide width variation, there are several key factors that affect the final dimension:

- Resist imperfections: including thickness, sensitivity, and age;
- Exposure change;
- Development: the development time and the age of the chemical;
- Etching.

Considering these manufacturing sources of width variation, it is expected that the average size of the fabricated waveguides is equal to 500nm , with a variation within the $+10/-30\text{nm}$ range.

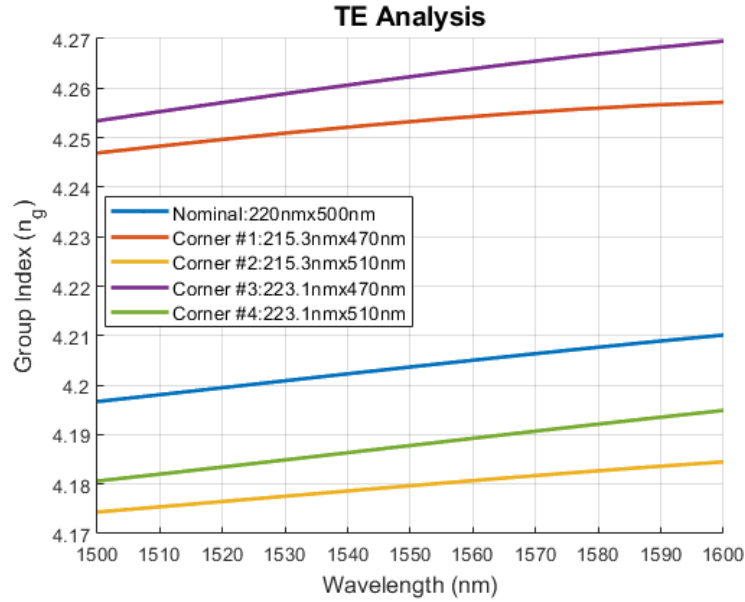


Fig. 17. Corner analysis of the Group Refractive index vs Wavelength

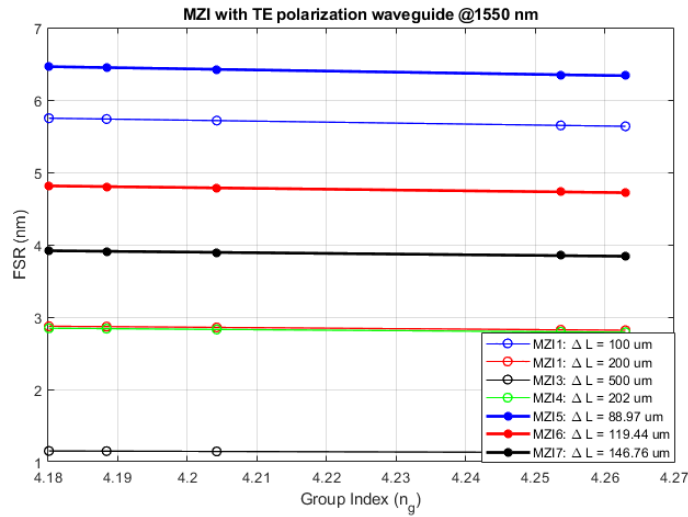


Fig. 18. Corner analysis of the FSR vs Wavelength

Figures 17 and 18 show the results of the corner analysis, which accounts for the combined effects of both width and thickness variability on the device performance.

8. Experimental Data Analysis

To estimate the performance of the designed devices, we extracted waveguide parameters from experimental results. Most importantly the group index of waveguides is found with an advanced curve fitting method. Before using this advanced curve fitting method, loop back structures are used to eliminate the effects of grating couplers from MZI transmission spectrum. As shown in

Fig 19, Loop back structure is consist of two grating couplers connected by a waveguide path. In this report, wavelength range with 10 dB variation in loss is used to avoid using noisy parts of transmission spectrum. Then, this loss is subtracted from the transmission spectrum of the MZIs. Finally, a calibrated transmission spectrum is obtained which is used in curve fitting to find fabricated waveguide parameters.

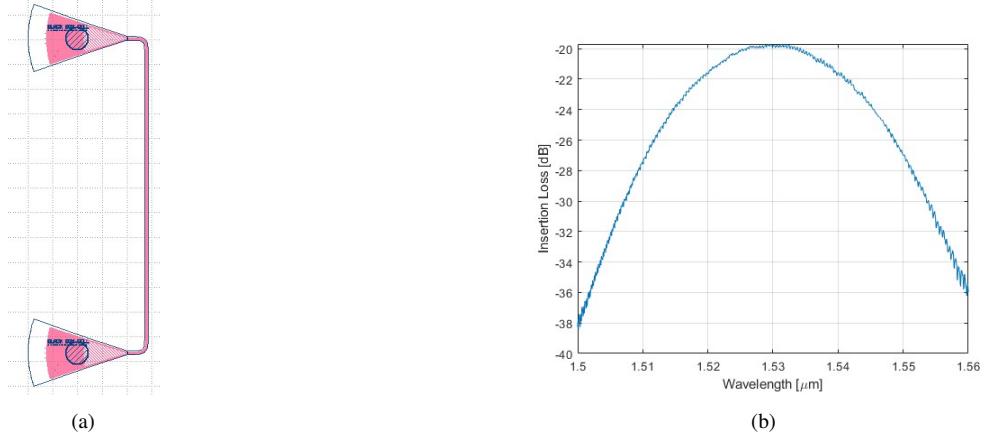


Fig. 19. Loop back structure for a) TE mode, b) measured insertion loss for TE mode corresponding to structure

The original(a) calibrated(b) and fitted (c) transmission spectrum of an MZI with waveguide cross section of $220 \times 500 \text{ nm}^2$ and path length difference of 88.97 m is shown in Fig. 20. Beside, the extracted group index for this MZI is plotted in Fig. 20 (d).

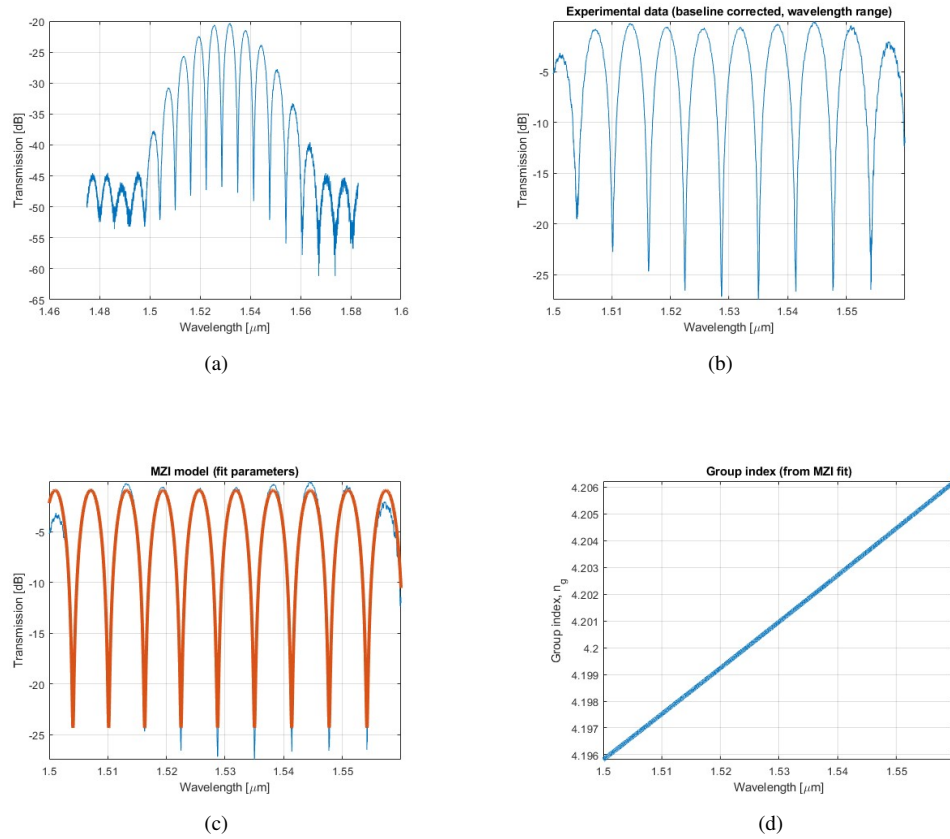


Fig. 20. (a)Original transmission spectrum (b) Calibrated spectrum (c) Fitted spectrum (d)Fitted group index for MZI with $\Delta L = 88.97 \mu\text{m}$.

Figure 21 summarizes the results of both simulations and experiments for different structures with various cross sections, path length differences, and polarizations. The group indices and FSRs are found at $1.55 \mu\text{m}$. This is a fairly good agreement between the simulation and the experimental results. The variability can be mainly attributed to the fitting error. Figure 22 shows the FSR change with respect to different length of MZI.

Cross-section(nm^2)	Mode	Device	$\Delta L(\mu\text{m})$	Group Index(Simulated)	FSR (nm)(Simulated)	Group Index(Experimental)	FSR (nm)(Experimental)	Error % FSR
500*220	TE	MZI1	100	4.181	5.746	4.1593	5.672	1.28785242
		MZI2	200		2.873	4.1775	2.8304	1.48277062
		MZI3	500.367		1.148	4.1837	1.1284	1.70731707
		MZI4	201.876		2.846	4.1745	2.8027	1.52143359
		MZI5	88.97		6.458	4.1937	6.316	2.19882317
		MZI6	146.76		3.915	4.2787	3.7458	4.32183908
		MZI7	119.44		4.811	4.1959	4.694	2.43192683

Fig. 21. Simulation and experimental results for the MZI with different parameters

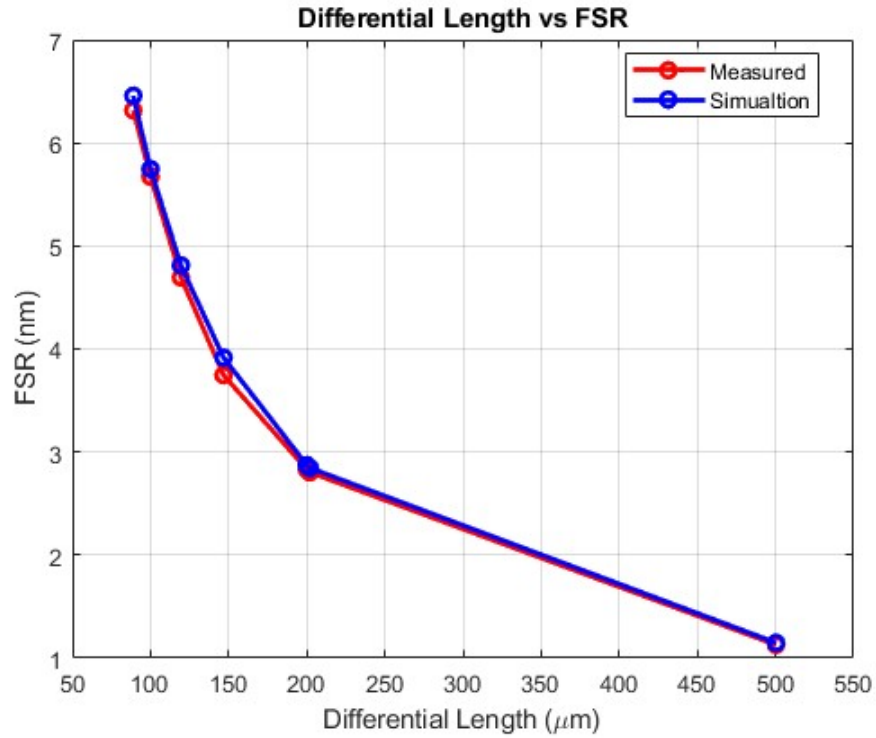


Fig. 22. FSR of measured data and simulated results for different differential lengths

9. Summary

This report described the steps regarding the design and fabrication and measurement of layout of MZIs, using Silicon photonics technology. The MZIs are designed with the waveguides in the strip configuration, with 220nm-height and 500nm-width. All the passive devices were fabricated using electron beam lithography, at the University of Washington facility. The devices were designed for TE modes, using Lumerical and Matlab for the design, and Klayout for the mask drawing. The group index and the FSR were the main parameters used for comparison between the designed and fabricated versions. The error analysis is done for FSR (simulated and measured). While processing the measured data, the laser alignment was noticed to be at 1530 nm (instead of 1550 nm) which can be a possible source of error.

10. Acknowledgment

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Omid Esmaeeli performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

References

1. H. Subbaraman et al, "Recent advances in silicon-based passive and active optical interconnects," *Opt. Express* **23**, 2487-2511 (2015).
2. K. Roberts et al, "High Capacity Transport 100G and Beyond," *Journal of Lightwave Technology*, vol.33, no.3, pp.563,578, Feb.1, 1 2015.
3. Y. Shoji et al, "Low-crosstalk 2x2 thermo-optic switch with silicon wire waveguides," *Opt. Express* **18**, 9071-9075 (2010).
4. T. Li, E. Willner and I. Kaminow, "Optical Fiber Telecommunications VA: Components and Subsystems," Academic Press, 2010.
5. L. Chrostowski and M. Hochberg, "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015.
6. Lumerical MODE software. <https://www.lumerical.com/tcad-products/mode/>.
7. Lumerical INTERCONNECT software. <https://www.lumerical.com/tcad-products/interconnect/>.
8. Components library, SiEPIC, University of Washington. <http://goo.gl/B9IRn8>.
9. KLayout software. <http://www.klayout.de/>.
10. R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," *J. Vacuum Sci. Technol. B* **29**, 06F309 (2011)
11. <http://siepic.ubc.ca/probestation>, using Python code developed by Michael Caverley.
12. Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", *Optics Express* Vol. 22, Issue 17, pp. 20652-20662 (2014)
13. PLC Connections, Columbus OH, USA www.plcconnections.com
14. Maple Leaf Photonics, Seattle WA, USA <http://mapleleafphotonics.com>